ATE MAKOCE

An Environmental Atlas

Land and Vegetation Change | 2015-2023

Wakpamni | Pine Ridge Indian Reservation

COMMUNITY LAND ASSESSMENT

Ate Makoce

An Environmental Atlas of Land and Vegetation Change

Prepared as a research and discussion resource for the Tribal landowner.

Environmental Data Cube for Oglala Lakota

INTRODUCTION

Reading the Land

This atlas brings together satellite observations, maps, and photographs to examine patterns in vegetation and land cover.

The maps are offered as a starting point for conversation, not as a replacement for knowledge held by people who know the land.

A satellite can help us see patterns across years. Understanding why those patterns occur requires local experience, field observations, and care.

ART AND PLACE

Art Alley

Photograph supplied for this atlas. Artwork, artist, and location credits require confirmation.

RESEARCH ETHICS

Our Land. Our Knowledge. Our Data.

Tribal Data Sovereignty recognizes the right of Tribal Nations to govern data about their lands, communities, and resources.

That includes decisions about how information is collected, interpreted, stored, shared, and used.

Satellite imagery is one source of information. Landowner experience, community knowledge, and Tribal governance are essential to how research findings are understood and shared.

GUIDE TO THE ATLAS

What the Maps Can Tell Us

These maps summarize observations of vegetation and land cover. Read each map alongside its original legend and classification.

Greenness is not the same as grazing condition. Lower greenness may be influenced by drought, seasonality, soil conditions, or other factors.

The facing pages offer interpretations and questions to discuss with the landowner. They are not definitive conclusions about land management.

PART III

The Research Atlas

Eight maps of land, vegetation, and change over time.

Each map is paired with an interpretation and a question for discussion.

PLATE 01

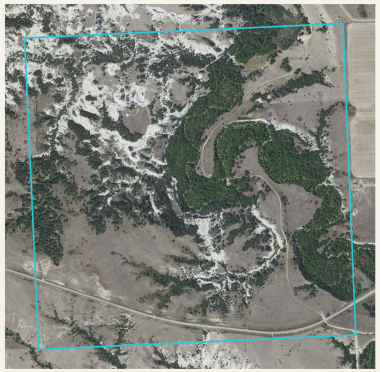
How Is My Land Doing?

ENVIRONMENTAL DATA CUBE FOR OGLALA LAKOTA

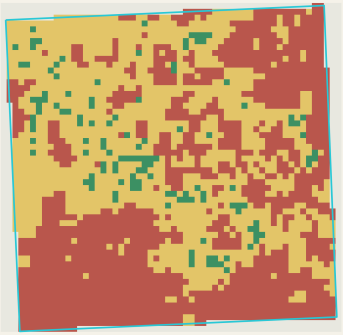
ATE_MACOKE | HOW IS MY LAND DOING?

Aerial photography, vegetation change, and land-cover information

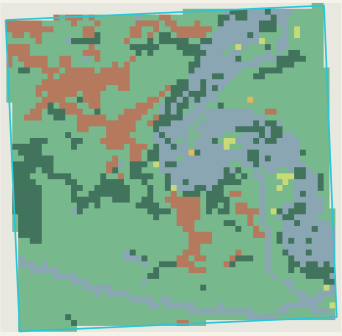
A / WHAT THE LAND LOOKS LIKE



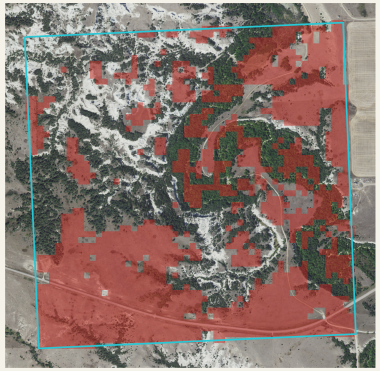
B / HOW VEGETATION CHANGED



C / WHAT KIND OF LAND IS IT?



D / WHERE SHOULD WE LOOK CLOSER?



VEGETATION CHANGE

Less green | Little change | Greener

LAND-COVER TYPES

Grassland / Pasture | Cultivated Crops | Forest / Shrubland

Hay / Alfalfa | Barren | Other Land

Red areas show reduced vegetation greenness from 2015 to 2023, not confirmed land degradation. Aerial imagery date may differ from the NDVI comparison years.

PLATE 01 | INTERPRETATION

How Is My Land Doing?

READING THE MAP

A combined view of several land observations. The different panels provide complementary evidence rather than a single measure of land condition.

FOR DISCUSSION

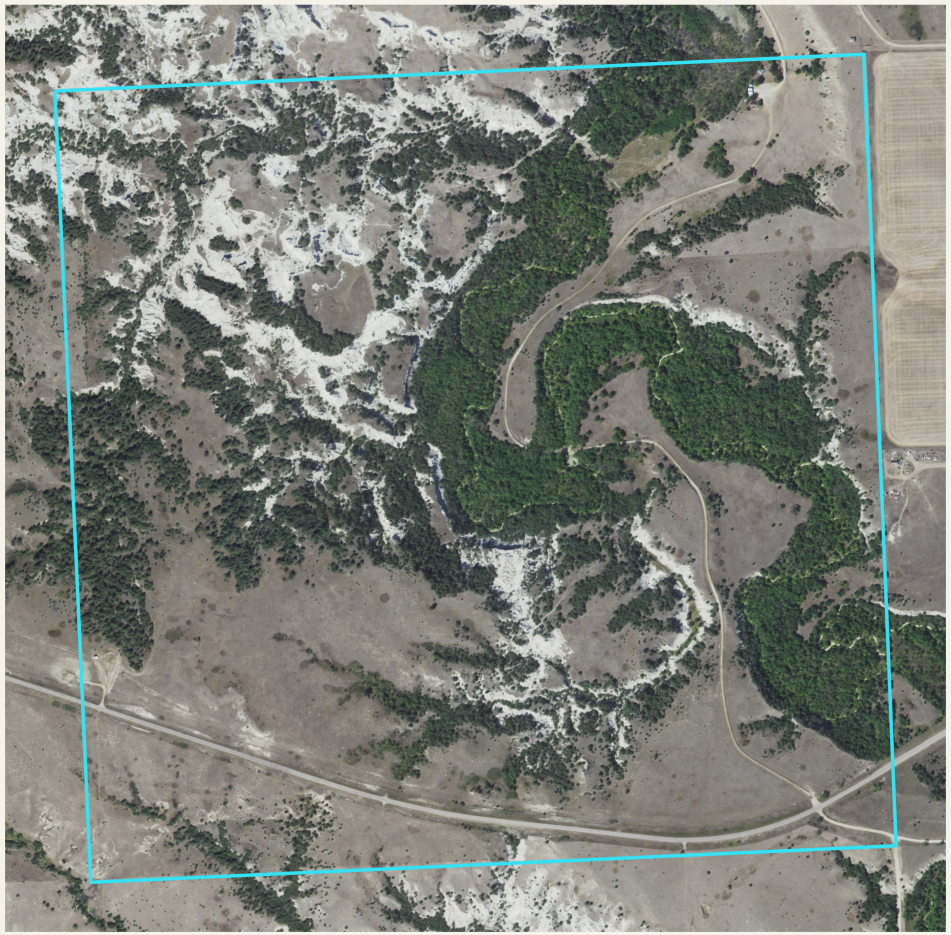
Which observations match what you have seen?

LANDOWNER OBSERVATIONS

PLATE 02

A View of the Land

ATE_MACOKE
A VIEW OF THE LAND



Aerial imagery: USGS / USDA NAIP | Cyan outline: Ate_Macoke parcel

PLATE 02 | INTERPRETATION

A View of the Land

READING THE MAP

Aerial imagery and parcel boundaries help place the research in its geographic setting.

FOR DISCUSSION

Which features and places do you recognize?

LANDOWNER OBSERVATIONS

PLATE 03

The Sampling Lattice

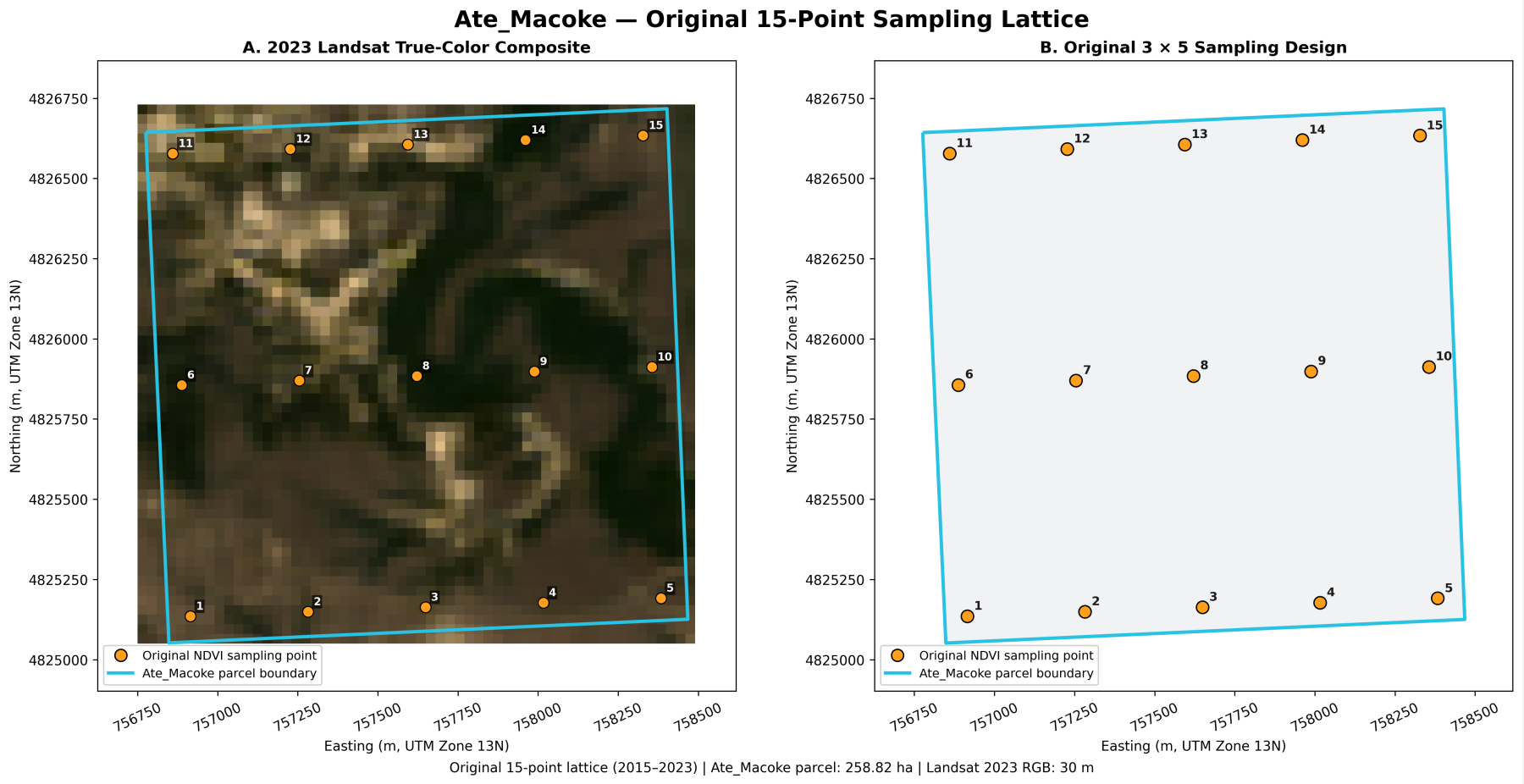


PLATE 03 | INTERPRETATION

The Sampling Lattice

READING THE MAP

The sampling locations organize the observations used to examine vegetation across the study area.

FOR DISCUSSION

Where might additional observations be useful?

LANDOWNER OBSERVATIONS

PLATE 04

Vegetation Through Time

PLATE 04 | INTERPRETATION

Vegetation Through Time

READING THE MAP

Vegetation greenness varies between years. Rainfall, temperature, seasonal timing, and land management may all contribute.

FOR DISCUSSION

Which years do you remember as unusually dry or wet?

LANDOWNER OBSERVATIONS

PLATE 05

Annual Vegetation Change

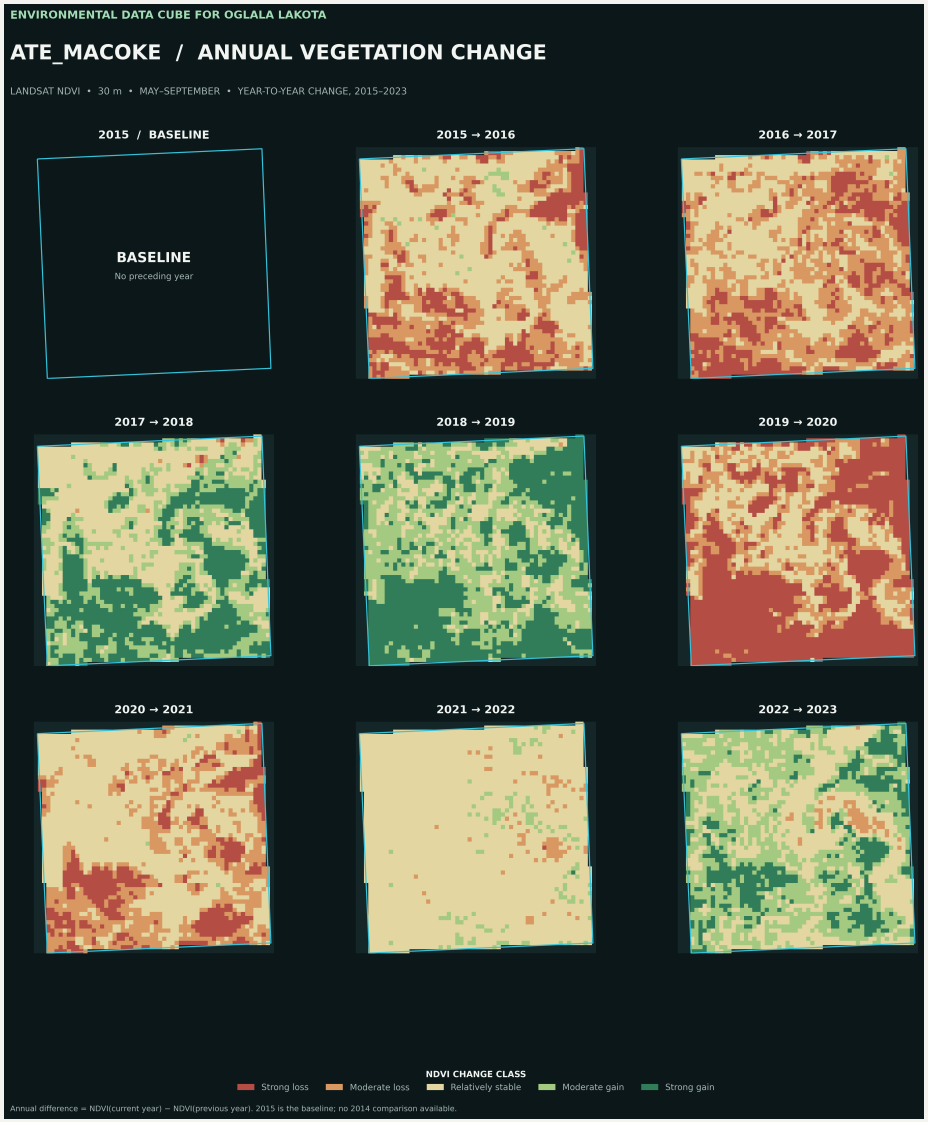


PLATE 05 | INTERPRETATION

Annual Vegetation Change

READING THE MAP

Comparisons across years help reveal patterns of increasing and decreasing vegetation greenness.

FOR DISCUSSION

Where do these changes agree with your experience?

LANDOWNER OBSERVATIONS

PLATE 06

Low Greenness History

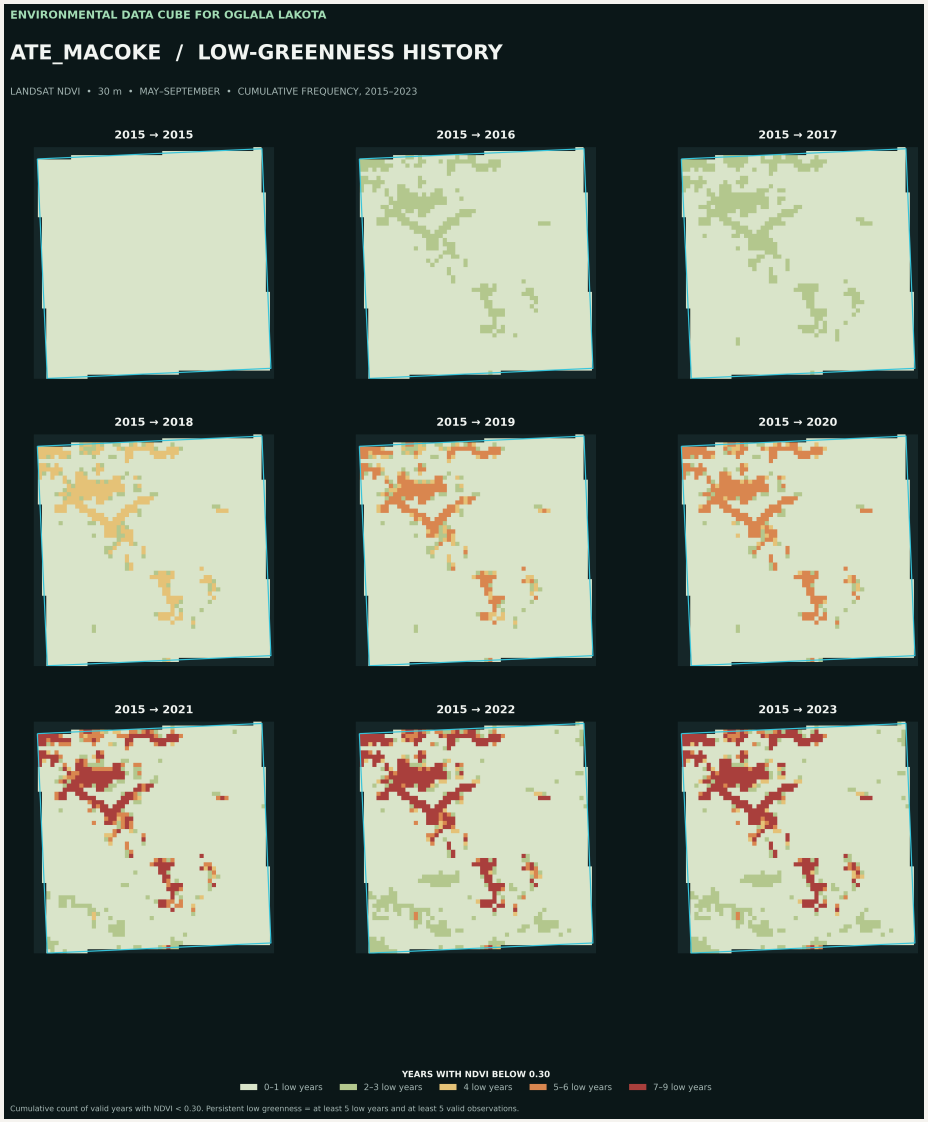


PLATE 06 | INTERPRETATION

Low Greenness History

READING THE MAP

Repeated lower greenness identifies areas for closer examination. It is not, by itself, evidence of overgrazing.

FOR DISCUSSION

What conditions might explain these areas?

LANDOWNER OBSERVATIONS

PLATE 07

Change and Persistence

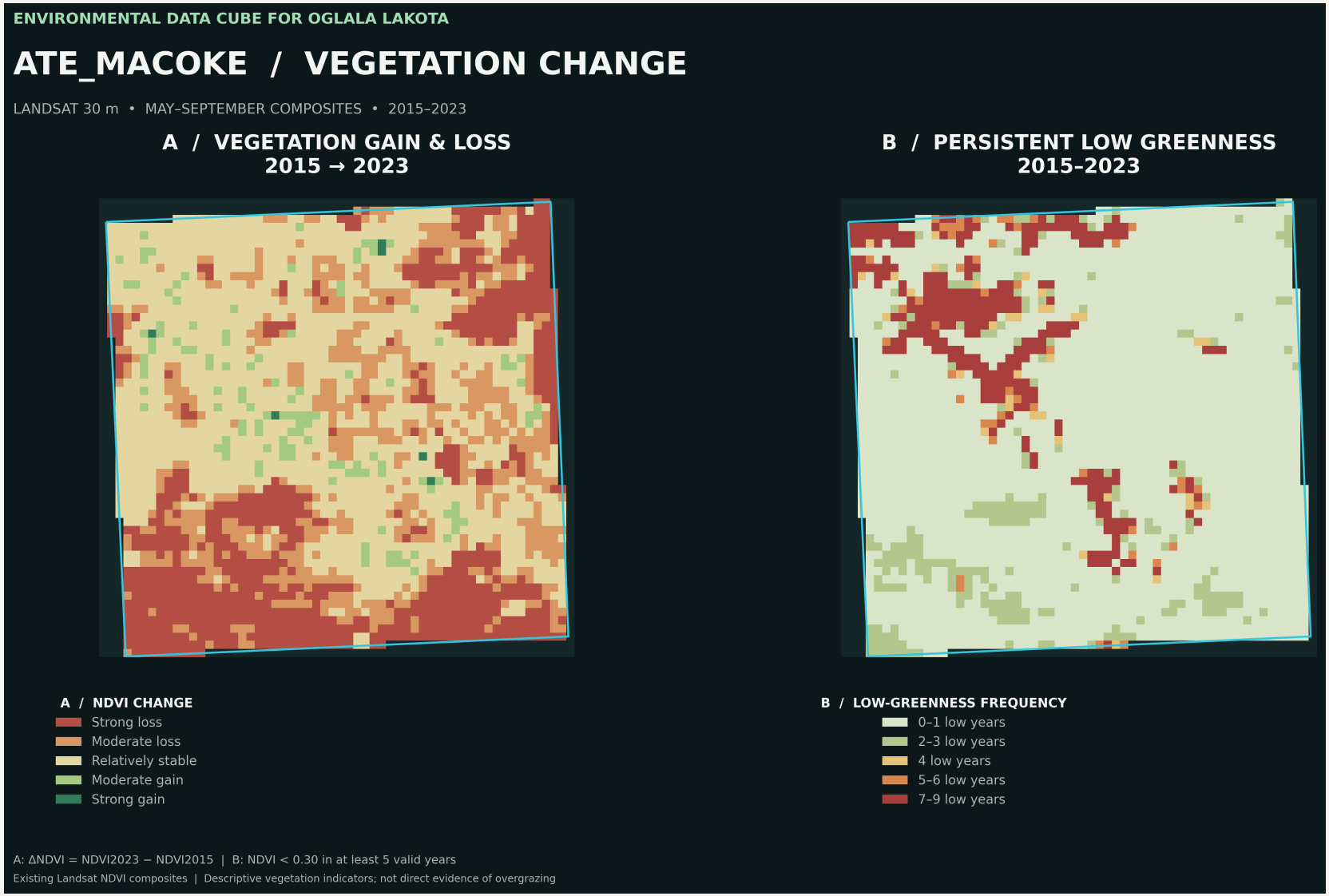


PLATE 07 | INTERPRETATION

Change and Persistence

READING THE MAP

Some vegetation patterns recur, while others change across the observation period.

FOR DISCUSSION

Which locations should be visited or discussed?

LANDOWNER OBSERVATIONS

PLATE 08

Vegetation and Land Cover

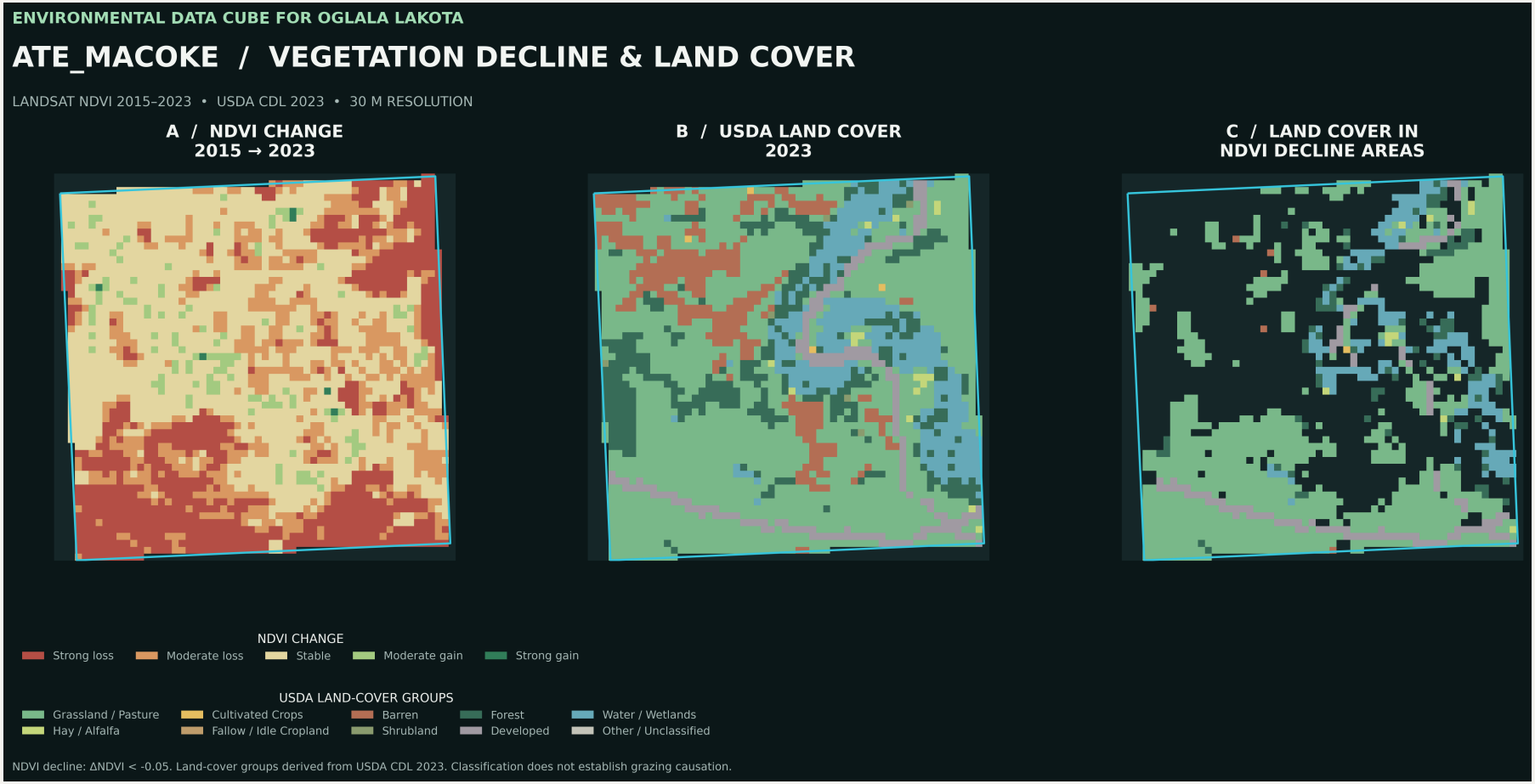


PLATE 08 | INTERPRETATION

Vegetation and Land Cover

READING THE MAP

Vegetation and land-cover observations can help guide questions about the landscape.

FOR DISCUSSION

What additional local knowledge would help?

LANDOWNER OBSERVATIONS

INTERPRETATION

What Remains to Be Learned

Satellite observations can reveal patterns that may deserve closer examination.

They cannot, by themselves, establish the cause of a change or determine whether grazing practices are sustainable.

Field observations, weather records, and knowledge of land use are needed to interpret the maps.

Decisions about further analysis and sharing should follow appropriate landowner and Tribal data-governance processes.

FIELD NOTES

Observations and Questions

What should we look at together next?

Ate Makoce

An Environmental Atlas of Land and Vegetation Change

A resource for observation, discussion, and continued learning about the land.